Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

PHYSICS **5054/42**

Paper 4 Alternative to Practical **October/November 2016**

MARK SCHEME

Maximum Mark: 30

Published

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1	(a)	(i)	64(cm^3)	B1
		(ii)	reading top of meniscus instead of bottom / parallax error explained	B1
	(b)	(i)	balance / scales	B1
		(ii)	find mass using tare / subtract mass of measuring cylinder from that of measuring cylinder + liquid	B1
			(density) = mass/volume	B1
	(c)		smaller value for density	B1

[Total: 6]

2	(a)	(i)	crocodile clips	B1
		(ii)	(close jaws) gently **or** use ratchet / thimble / spindle **or** until wheel slips	B1
			repeat at different places / positions (and average)	B1
		(iii)	0.055796(using π button) / 0.055768(using 3.14) / 0.055818 (using 22 / 7)	C1
			0.056	A1
	(b)	(i)	axes labelled quantity and unit **and** axes correct way round	B1
			scales linear, not awkward, start from (0,0)	B1
			points plotted accurately	B1
			best-fit straight line drawn	B1
		(ii)	large triangle or any other indication of chosen points shown on graph	B1
			93 ± 2 accept numbers rounding down to 95 and up to 91 not accept fractional values	B1
		(iii)	candidate's **(a)(iii)** ÷ 200×**(b)(ii)**	C1
			answer correct (must be checked) in standard form	A1

[Total: 13]

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3	**(a)**	orange, orange	B1
		brown	B1
	(b)	power supply, fixed resistor and diode (any orientation) in series	B1
		ammeter in series (with diode and resistor)	B1
		voltmeter across diode	B1
	(c)	variable power supply add rheostat / variable resistor add resistor (in series) / use different values of resistor add cells / batteries	B1
	(d)	reverse power supply reverse diode / it	B1
			[Total: 7]
4	**(a)**	27	B1
	(b) **(i)**	range 100–140	C1
		range 110–130	A1
	(ii)	4 × their **(b)(i)**	B1
			[Total: 4]

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